

CHG 569

Environmental Pollution Engineering

Module X

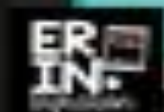
- The next pollution is here

Environmental

impact of technology



LOVE



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Digital Pollution



The digital revolution has evolved through four phases, beginning with big, expensive, standalone computers, and progressing to today's digital world in which small, inexpensive digital devices are everywhere

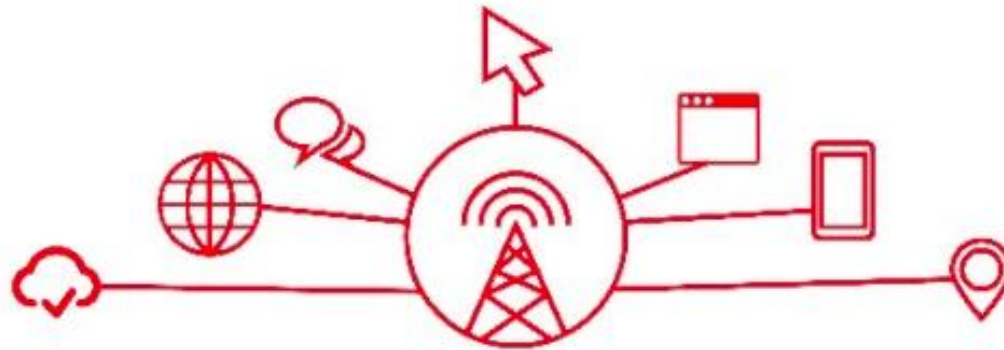
Expired	Tired	Uninspired	Desired
Data processing	Personal computing	Network computing	Cloud computing
Big corporate and government computers	Desktop computers	Notebook computers	iPhone, Droid X, and iPad-like devices
Custom applications	Standalone applications	Monolithic software suites	Handheld apps and cloud-based apps
CB radios	Dial-up Internet access	Cable, satellite Internet access	3G, Wi-Fi Internet access
ARPANET	AOL and CompuServe	The Web and virtual worlds	Social media
Arcade games	2-D action games	3-D multiplayer games	Touch screen microgames for handheld devices

Why digital ?

The Revolutions have been great turning points in history and have been of immense benefit to man, hence the term “**REVOLUTION**”

- 1st Revolution **mechanical revolution-** hand production to machines, mechanization driven by steam power
- 2nd Revolution- discovery of electricity lead to the **electical revolution,** with inventions like electric lighting, radio, and phone.
- 3rd Revolution-computers and programmable control lead to the **automation revolution,** semiconductors, mainframe computing, personal computing, and the Internet. Analog equipment became digital (TVs of old were tuned with an antenna (analog) now replaced by an Internet-connected tablet that lets you stream movies (digital).
- 4th Revolution characterized by the application of information and communication technologies mainly to industry but also to the rest of societies' activities and builds on the 3rd revolution. Industry 4.0 is actually the digitalization of the manufacturing environment with Big Data, Internet of Things, Data Analytic, Smart systems, simply put, there is an enormous increase in the use of digital devices (, thus:

With the internet came an information revolution, this information is usually represented as data, which in turn is represented as discrete digits (use of numbers – 0 and 1). Digital electronics are used in devices to store, process and transmit data.



THE DIGITAL REVOLUTION IS UPON US

Telecom companies are at the heart of the digital revolution. Why? The telecom industry provides the building blocks – access, interconnectivity, and applications – to create the foundation for global digitization.

Today's world runs on DIGIT – Data is the new oil







**What is the
environmental
impact of digital
technology?**

wedia

- **Carbon footprint**
- **Water footprint**

Digital Pollution: What is it?

- environmental pollution produced by digital tools during 'take-make-waste'
- the greenhouse emissions – carbon footprint - behind our digital activity – The humongous amount of energy required From the consumption of data centres to transmission networks to our handheld devices these are the pollution drivers

Digital Pollution drivers

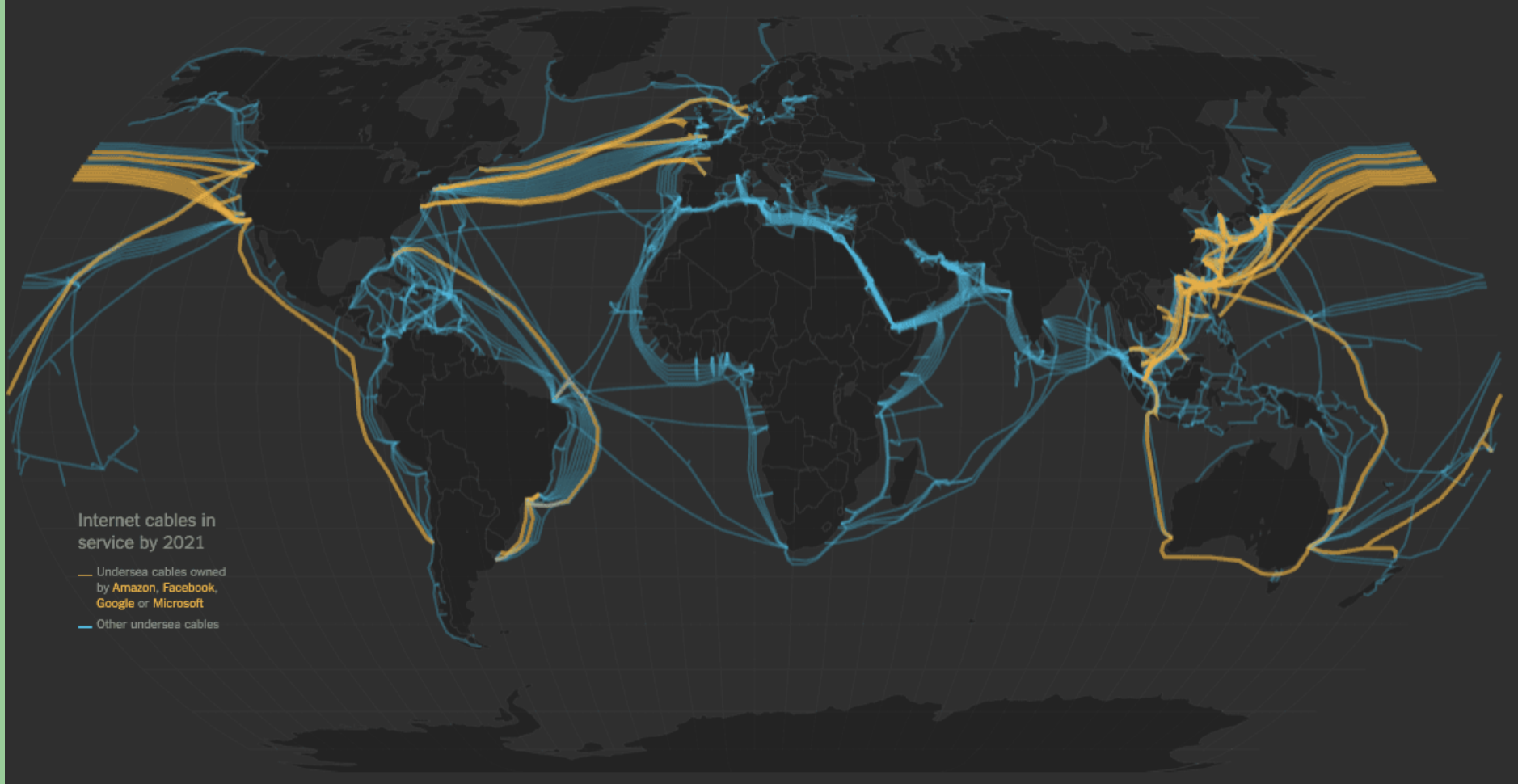
- **Data Centers** – store & host webpages
- **Access networks** – wiring and antennas that carry data
- **End-user devices** – computers, tablets, laptop and routers

A data centre





‘People think that data is in the cloud, but it’s not. It’s in the ocean.’



How the Internet Travels Across Oceans. Source: [New York Times](#)

Digital Pollution: Causes

Digital technology means using energy, it is therefore an environmental issue

- **Mining for components** – smart phones, computers, etc. have metal components that need to be mined, and mining causes water pollution, dangers to human health, and destroys ecosystem
- **Excessive energy consumption** for manufacturing, transportation, and use - According to a study carried out in 2019 by a digital expert, it represents nearly 3.8% of global greenhouse gas emissions **the ranking of impact sources (in decreasing order of importance) is as follows:**
 1. Equipment manufacturing ;
 2. Electricity consumption of the equipments ;
 3. Network power consumption ;
 4. Data centres' power consumption ;
 5. Manufacture of network equipments ;
 6. Manufacture of equipment hosted by IT centres (*servers, etc.*)
- **Electronic waste** – rapid turnover of electronic devices due to rapid growth in technology – new devices with superior features making older devices obsolete => increase in e-waste generation, pollution and raw material depletion. The consumption and production of electronic devices follow the linear model of take-make-dispose [not good for the environment]
 - E-waste in landfills - hazardous substances in e-waste leach into the ground and release toxic gases

The Environmental Impact of Digital Technologies and Data

Manufacturing means resource depletion

JUL
2022

ESSENTIAL DIGITAL HEADLINES

OVERVIEW OF THE ADOPTION AND USE OF CONNECTED DEVICES AND SERVICES



TOTAL
POPULATION



we
are
social

7.98
BILLION

URBANISATION

57.0%

UNIQUE MOBILE
PHONE USERS



5.34
BILLION

vs. POPULATION

66.9%

INTERNET
USERS



5.03
BILLION

vs. POPULATION

63.1%

ACTIVE SOCIAL
MEDIA USERS



4.70
BILLION

vs. POPULATION

59.0%

SOURCES: UNITED NATIONS; U.S. CENSUS BUREAU; GOVERNMENT BODIES; GSMA INTELLIGENCE; ITU; GWI; EUROSTAT; CNNIC; APJII; CIA WORLD FACTBOOK; COMPANY ADVERTISING RESOURCES AND EARNINGS REPORTS; OECD; TECHRASA; KEPIOS ANALYSIS. **ADVISORY:** SOCIAL MEDIA USERS MAY NOT REPRESENT UNIQUE INDIVIDUALS. **COMPARABILITY:** SOURCE AND BASE CHANGES. THE U.N. REVISED ITS POPULATION DATA SINCE OUR PREVIOUS REPORT, WHICH MAY AFFECT ALL VALUES THAT COMPARE DIGITAL ACTIVITY TO POPULATION, AND MAY RESULT IN APPARENT DECREASES IN DIGITAL ADOPTION. HOWEVER, WE ADVISE CAUTION WHEN INTERPRETING ANY CHANGES IN THESE COMPARATIVE FIGURES, BECAUSE ANY SUCH CHANGE MAY BE SOLELY THE RESULT OF REVISIONS TO POPULATION DATA.

we
are
social

Hootsuite

A BREAKDOWN OF THE CRITICAL METALS IN A SMARTPHONE

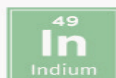
Some vital metals used to build these devices are considered at risk due to geological scarcity, geopolitical issues or trade policy.

This infographic details the critical metals that you carry in your pocket.

ALKALI METAL ALKALINE EARTH TRANSITION METAL BASIC METAL LANTHANOID

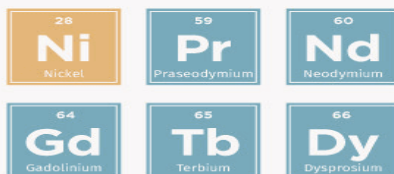
TOUCH SCREEN

It contains a thin layer of **indium** tin oxide, highly conductive and transparent, allowing the screen to function as a touch screen.



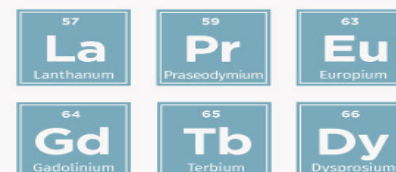
MICROPHONE, SPEAKERS, VIBRATION UNIT

Nickel is used in the microphone diaphragm (that vibrates in response to sound waves). Alloys containing **neodymium**, **praseodymium** and **gadolinium** are used in the magnets contained in the speaker and microphone. **Neodymium**, **terbium** and **dysprosium** are used in the vibration unit.



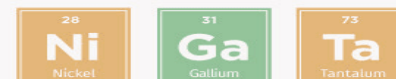
DISPLAY

The display contains several **rare earth elements**. Small quantities are used to produce the colors on the liquid crystal display. Some give the screen its glow.



ELECTRONICS

Nickel is used in electrical connections. **Gallium** is used in semiconductors. **Tantalum** is the major component of micro capacitors, used for filtering and frequency tuning.



CASING

Nickel reduces electromagnetic interference. **Magnesium** alloys are superior at electromagnetic interference (EMI) shielding.



BATTERY

The majority of smartphones use **lithium-ion** batteries.

Source: University of Birmingham

Conflict minerals clampdown

The *Securities and Exchange Commission* has ruled that U.S.-listed manufacturers such as Apple and Boeing must scrutinise the sources of four metals to make sure they don't help fund human rights abuses

ANATOMY OF A SMART PHONE

50

Sn
Tin

Tin aka stannum (Latin)

Silvery, malleable metal that does not easily oxidize in air

Source: Cassiterite

Use: Circuit board *solder* and battery *anodes*

73

Ta
Tantalum

Tantalum from Tantalus

(Greek mythological figure)

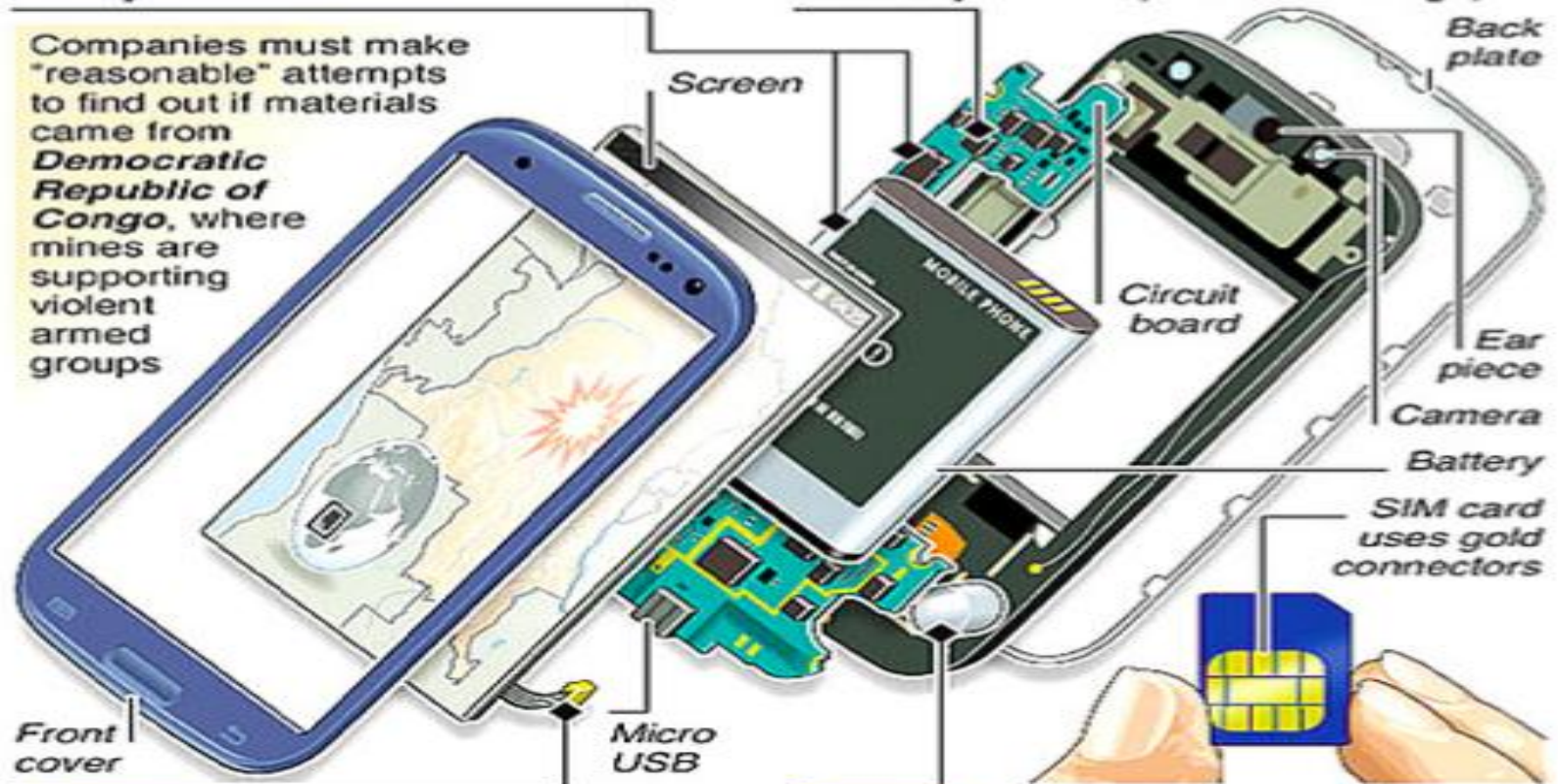
Rare, hard, lustrous metal.

Highly corrosion resistant

Source: Columbite-tantalite (Coltan)

Use: *Capacitors* (electrical storage)

Companies must make "reasonable" attempts to find out if materials came from **Democratic Republic of Congo**, where mines are supporting violent armed groups



79

Au
Gold

Gold aka aurum (Latin)

Dense, soft, malleable metal

Source: Nuggets or grains in rock and alluvial deposits

Use: *Connectors* – does not corrode in air like silver and copper

74

W
Tungsten

Tungsten from tung sten

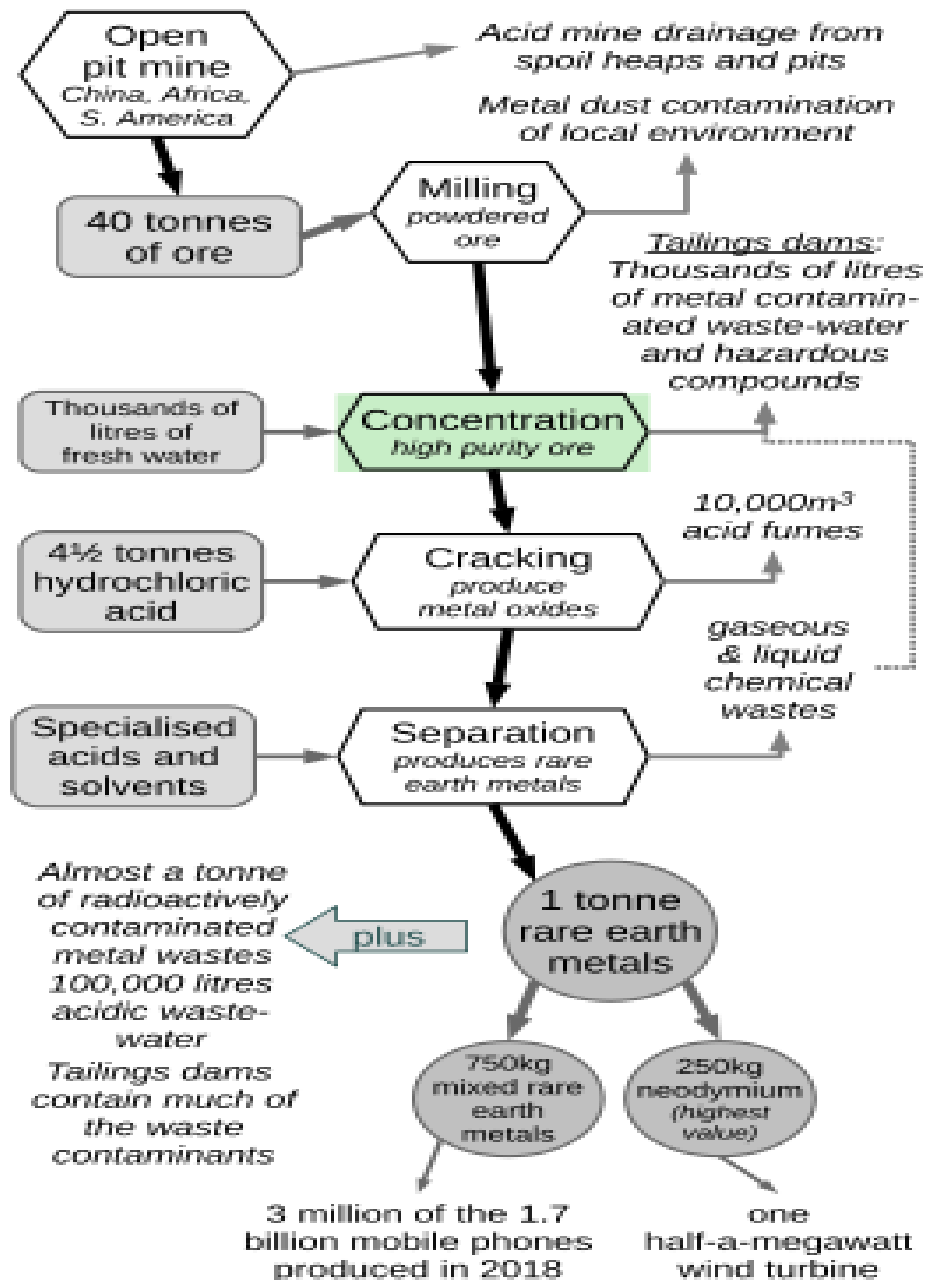
(Swedish) meaning "heavy stone"; aka **wolfram**

(German). Hard, rare metal

Source: Wolframite

Use: *Vibration motor*

Producing rare earth metals for digital & renewable energy technologies





*Rare earth metals refinery tailings lake,
China*

So how sustainable is the digital revolution



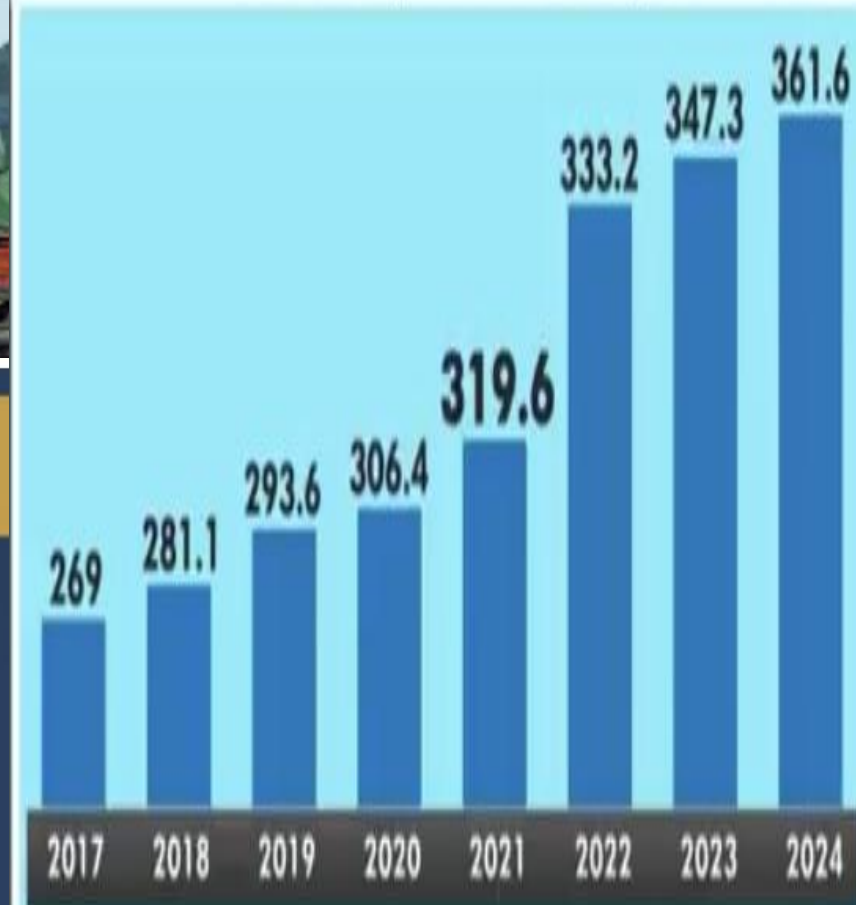
The world is choking on digital pollution



HOW BAD IS EMAIL FOR THE ENVIRONMENT?



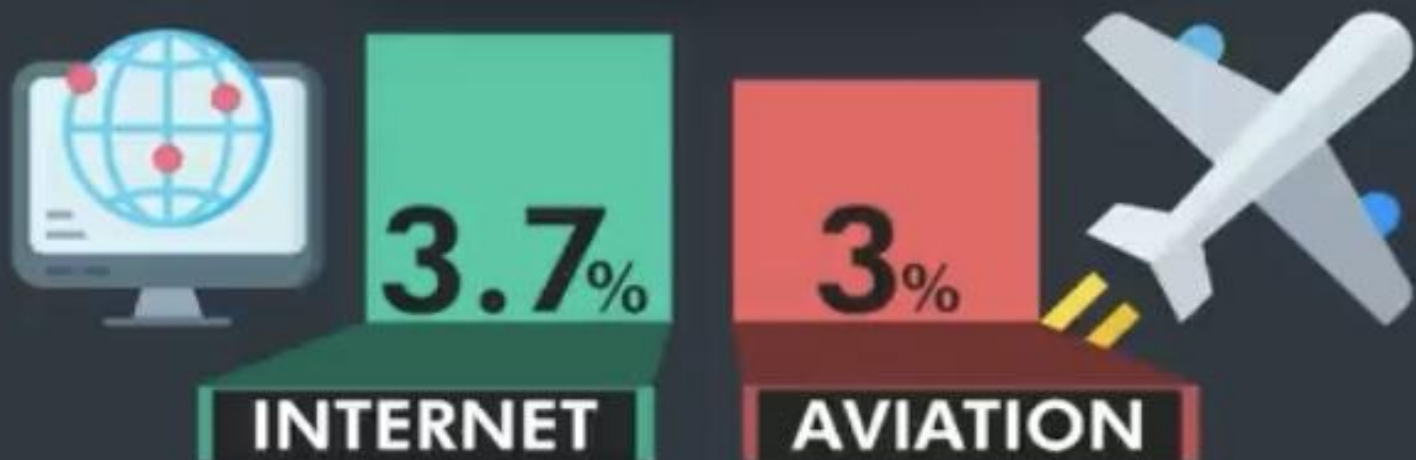
NUMBER OF SENT AND RECEIVED E-MAILS PER DAY (IN BILLIONS)



**ONLINE VIDEOS
RESPONSIBLE FOR OVER
300 MN TONNES
OF CO₂ EVERY YEAR**



SIZING UP CARBON FOOTPRINTS

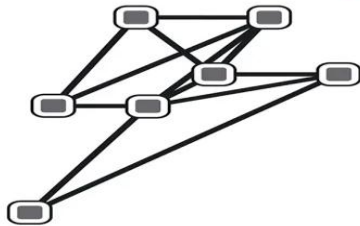




digital pollution

With 4% of global GHG emissions, digital technology pollutes more than the aviation industry

Source: The Shift Project, 2019

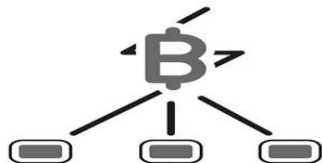


Over 7000 data centers use around 2% of global electricity

Source: Fortune, 2019

In 2018, online video streaming released more than 300 million tonnes of CO2 which is equivalent to the total annual GHG emissions of Spain

Source: The Shift Project, 2019



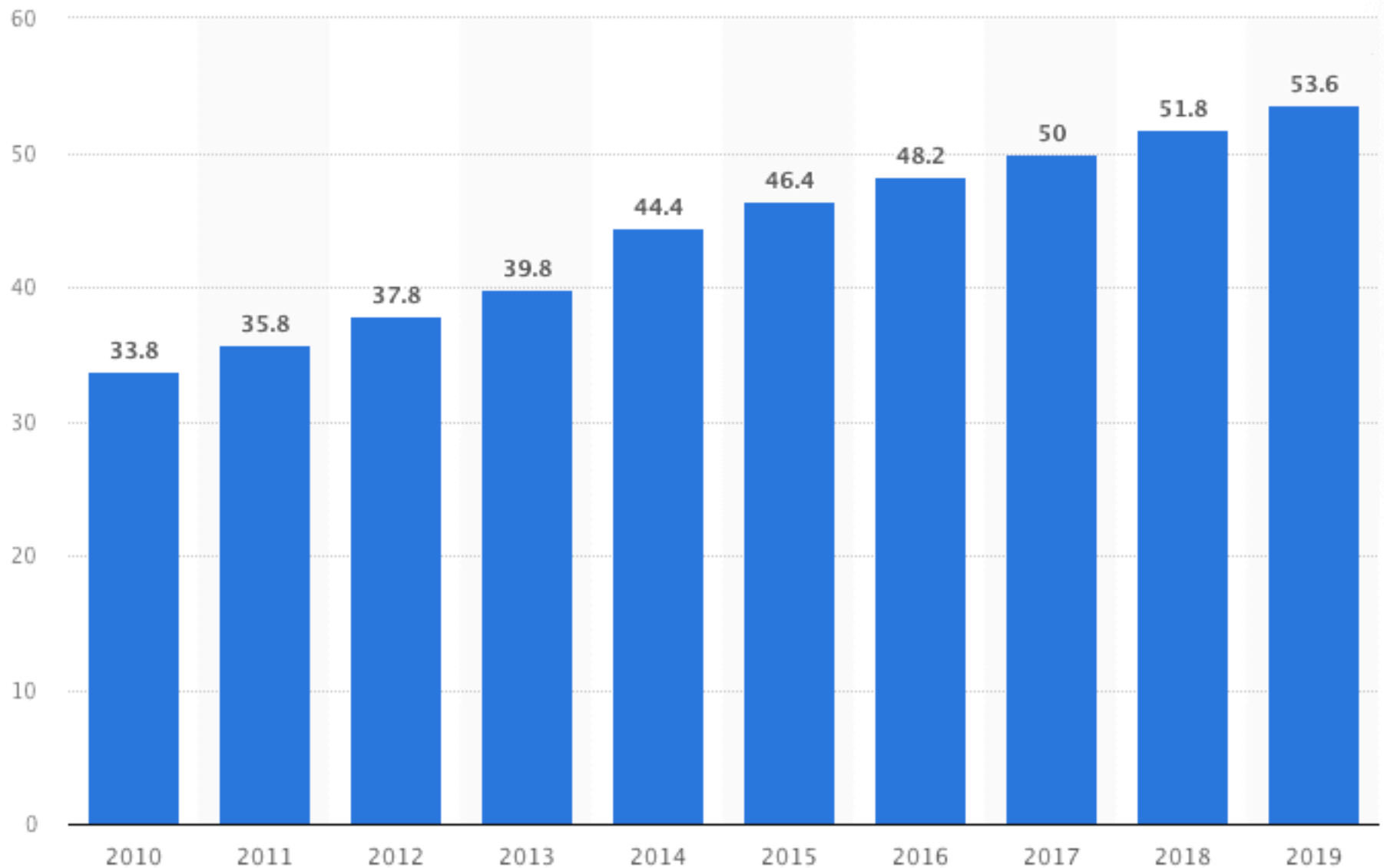
Bitcoin consumes 64.15 TWh of electricity per year, which exceeds the energy consumption of whole countries like Chile, Switzerland, New Zealand, or Bangladesh.

Source: The Verge, 2019

53 million tons of e-waste in 2019
(smartphones, computers, TV)

Only 20% of e-waste is collected and recycled. 80% end up in landfills





In 2019, the world generated **53.6 million metric tons of e-waste**, and only 17.4% of this was officially documented as properly collected and recycled.

Mitigating measures

- it's all about reducing energy usage

Mitigating Resource depletion & Waste generation

- Only replace electronic devices if really necessary, this helps to **reduce consumption of digital devices**
- Encourage **recycling of electronic devices** by devices made from recycle material

Mitigating energy consumption by efficient use of devices

- Reduce file size. ...
- Prefer download and reduce video streaming. ...
- Store the files on your hard drive. ...
- Subscribe only to newsletters that interest you. ...
- Delete junk files and unused apps. ...
- Turn off any devices you are not using.
- Reduce screen brightness

So how sustainable is the digital revolution



- Phone calls vs. emails
- Face-to-face meetings vs. zoom
- Physical shopping vs. online shopping
- Download vs. streaming
- Paper books vs. e-reader
- File cabinet vs. cloud storage

What of the water footprint?



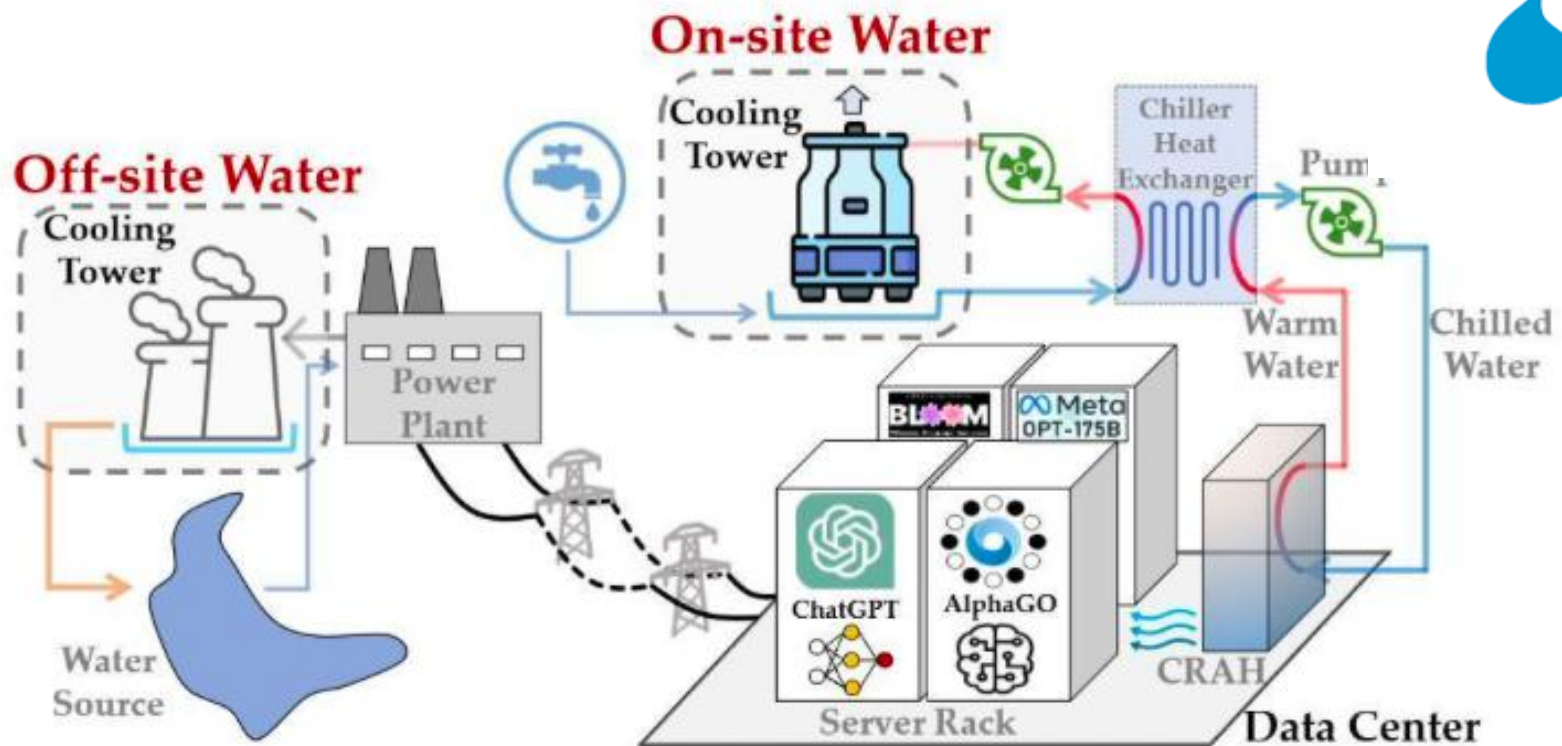
Considering AI models, e,g GPT-3

Training in a state-of-the-art U.S. data centre could **consume up to 700,000 litres of freshwater** – a great concern due to **freshwater scarcity**.

How?

- Direct consumption, on-site, for cooling servers
- Indirect consumption, off-site, for electricity generation

Water footprint of the digital revolution?



Digital Pollution

- **Ecological(resource /waste) footprint**
- **Carbon footprint**
- **Water footprint**